Dark Matter in the Universe

Answer key

Conventions used throughout

• **Orbit equation:** gravity is the centripetal force, $GMm/r^2 = mv^2/r$. The orbiting object's mass cancels, giving $v = \sqrt{(GM/r)}$ and $M = v^2r/G$, where M is the mass inside the orbit.

• **Constants:** $G = 6.67 \times 10^{-11}$ N·m^2/kg^2. Luminous mass of UGC 11748: 1.54×10^{41} kg.

• **Video questions:** the answers match the answer line printed on the worksheet. Every step keeps full precision; only final answers are rounded.

VIDEO

Multiple choice 1–4

Watch the video on Dark Matter and answer the multiple choice questions below while you are watching.

For each question, circle the letter corresponding to the correct answer: (graphs A–D are shown below the questions)

1. Which of the above graphs best shows the velocity versus orbital radius relationship for planets orbiting the Sun? a) A b) B c) C d) D

2. Which of the above graphs best shows the **V** versus **r** relationship observed for stars orbiting the centre of a galaxy? a) A b) B c) C d) D

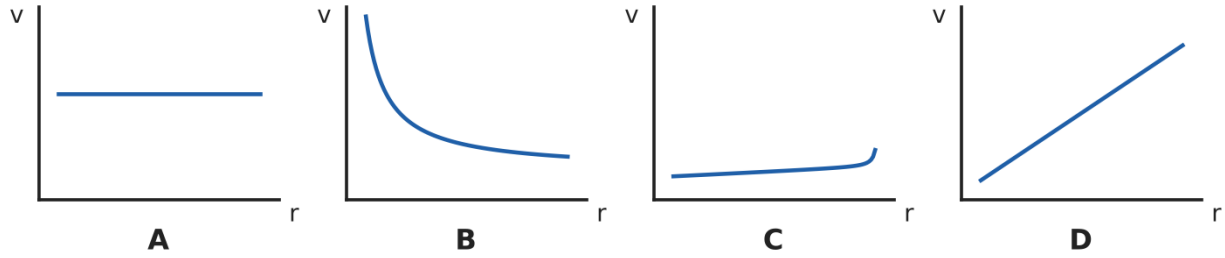
3. When astronomers measure the mass of the galaxy Triangulum using the Brightness Method the result they get is much less than when they measure the mass using the Orbital Method. This fact can best be explained by the fact that

   a) the stars in Triangulum are made of lighter elements than those in the Sun. b) the Orbital Method underestimates the mass of Triangulum. c) Triangulum contains a large amount of unseen mass. d) Triangulum contains only mass that emits visible light.

4. Observations that indicate the presence of dark matter have been made in

   a) only the Andromeda and Triangulum galaxies. b) only the Andromeda, Triangulum, and Milky Way galaxies. c) every galaxy that has been examined for dark matter. d) many, but not all, galaxies that have been examined for dark matter.

Graphs A–D (redrawn from the worksheet)



VIDEO (CONTINUED)

Multiple choice 5–8

5. Evidence for dark matter comes from observations of

   a) the orbit of the Moon around Earth and gravitational lensing. b) the orbits of stars and gravitational lensing. c) the orbit of the Moon around Earth and the orbits of stars. d) the orbits of stars, the orbit of the Moon around Earth, and gravitational lensing.

6. Dark matter is called “dark” because it

   a) only emits high-energy radiation such as X-rays and gamma rays. b) only emits low-energy radiation such as microwaves and radio waves. c) reflects light but does not emit other radiation like stars do. d) does not emit or reflect any type of radiation or light.

7. Most physicists think that most dark matter is made of

   a) WIMPs or axions. b) brown dwarf stars. c) black holes or planets. d) stars like the Sun.

8. Which of the following is true?

   a) Physicists know exactly what dark matter is made of. b) Physicists have no idea what dark matter is made of. c) Only some physicists know what dark matter is made of. d) Physicists have some ideas about dark matter, which they are currently testing by experiments.

#	Answer	Why
1	b) B	For planets almost all the mass is in the Sun, so v = √(GM/r): speed drops as r grows.
2	a) A	Stars far from a galaxy's centre move at about the same speed at every radius: a flat line.
3	c) Triangulum contains a large amount of unseen mass.	The orbits feel more mass than the light shows, so some mass gives off no light.
4	c) every galaxy that has been examined for dark matter.	As stated in the video.
5	b) the orbits of stars and gravitational lensing.	The Moon's orbit is explained by Earth's ordinary mass, so it isn't evidence.
6	d) does not emit or reflect any type of radiation or light.	That is why we can't see it.
7	a) WIMPs or axions.	Unknown particles; ordinary objects (stars, planets) would give off or block light we could detect.
8	d) Physicists have some ideas about dark matter, which they are currently testing by experiments.	No one knows yet what it is.

FINAL ANSWER

1 B, 2 A, 3 C, 4 C, 5 B, 6 D, 7 A, 8 D

SUMMARY

Summary of the physics

1. What is the “orbital method”? Specifically, what does it measure and how?
2. What is the “brightness method”?
3. When comparing the results of the orbital and the brightness method, what was found? How has the discrepancy been explained?

1. What is the “orbital method”?

It finds the mass of a galaxy from how fast its stars orbit. Astronomers measure each star’s orbital speed v (from the Doppler shift of its light) and its distance r from the centre. Gravity is the centripetal force, so the mass inside that orbit is:

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \quad \Rightarrow \quad M = \frac{v^2 r}{G}$$

2. What is the “brightness method”?

It finds the mass from the light. Astronomers add up how much light the galaxy gives off and, knowing how bright stars of a given mass are, convert the total light into a total mass of stars (the luminous mass).

3. What was found? How has the discrepancy been explained?

The orbital method gives a much larger mass than the brightness method, and the difference grows farther from the centre. The explanation is dark matter: matter that has mass and gravity but gives off and reflects no light, spread out well beyond the visible stars.

FINAL ANSWER

1. Mass from orbits: measure v and r, then $M = v^2 r/G$

2. Mass from the total light given off by the stars

3. Orbits show much more mass than the light does; the extra is unseen dark matter

QUESTION 1

Mass of the Sun from Mars' orbit

1. Mars orbits the Sun in uniform circular motion. The radius of Mars' orbit is 2.28 x 10^{11} m and its orbital speed is 2.41 x 10^{4} m/s.

   a. Draw the free-body diagram for Mars and use it to derive an expression for the mass of the Sun in terms of Mars' orbital speed, the radius of its orbit, and the universal gravitational constant.

   b. Use the expression derived in part a) to determine the mass of the Sun.

a. Draw the free-body diagram for Mars and derive an expression for the mass of the Sun.

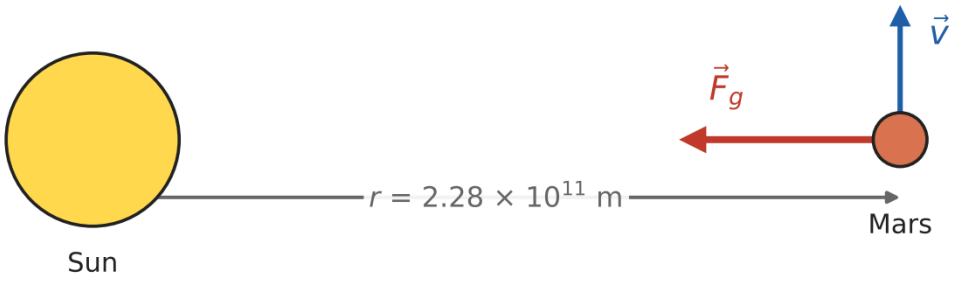
The only force on Mars is the Sun's gravity, pointing toward the Sun (the centre of the orbit). So it is the centripetal force:

$$F_c = F_g: \quad \frac{m_{Mars}v^2}{r} = \frac{Gm_{Sun}m_{Mars}}{r^2}$$

Divide both sides by m_{Mars} and multiply by r^2: $v^2 r = Gm_{Sun}$

Divide both sides by G: $m_{Sun} = \frac{v^2 r}{G}$

FBD of Mars



b. Use the expression derived in part a) to determine the mass of the Sun.

$$m_{Sun} = \frac{(2.41 \times 10^4)^2(2.28 \times 10^{11})}{6.67 \times 10^{-11}} = 1.9854 \times 10^{30} \text{ kg}$$

FINAL ANSWER

a. $m_{Sun} = v^2r/G$

b. $m_{Sun} = 1.99 \times 10^{30}$ kg

QUESTION 2

Orbital speed of the planets

2. The plot below relates the orbital speed of the planets to the radius of their orbits.

a. What is the force that keeps the planets in their orbits?

b. Why do the “outer” planets travel slower than the “inner” planets?

c. Rearrange your answer to 1a) to find the equation for the graph above.

a.

The gravitational force of the Sun on each planet.

b.

Gravity gets weaker with distance ($\propto 1/r^2$). A planet farther out feels a weaker pull toward the Sun, so it needs less centripetal force and moves more slowly: $v = \sqrt{(GM/r)}$ gets smaller as r gets bigger.

c.

Start from $m_{Sun} = v^2 r/G$ and solve for v:

Multiply by G and divide by r: $v^2 = \frac{Gm_{Sun}}{r}$

Square root both sides: $v = \sqrt{\frac{Gm_{Sun}}{r}}$

v is proportional to $1/\sqrt{r}$, which gives the falling curve in the plot.

FINAL ANSWER

a. The Sun's gravity

b. Gravity is weaker farther out, so less centripetal force and a slower orbit

c. $v = \sqrt{(Gm_{Sun}/r)}$

QUESTION 3

The galaxy UGC 11748

3. Astronomers have analyzed the stars in the galaxy UGC 11748. They found that most of the stars lie within a radius r = 1.64 x 10^{20} m and that the total mass within this radius is 1.54 x 10^{41} kg, or 77.4 billion times the mass of the Sun. It is expected that the stars that lie outside this radius will orbit in the same way that planets orbit the Sun. The following chart contains the data for various orbital radii and the corresponding measured speed of the stars located in the outer regions of UGC 11748. Fill in the following chart by performing the following steps:

a. For each orbital radius, calculate the speed expected if the only mass is the luminous mass of 1.54 x 10^{41} kg. Record your answers in the "Calculated speed" column. **(Show a sample calculation).**

b. Use the **measured** speeds to calculate the mass of the galaxy contained within each orbital radius. Record your answers in the "Gravitational mass" column. **(Show a sample calculation)**.

c. For each orbital radius, calculate the difference between the gravitational mass within this radius and the total mass of the stars (The total mass of the stars is 1.54 x 10^{41} kg based on the brightness method). Represent this difference as a percentage of the gravitational mass within the orbital radius. Record your answers in the "Missing Mass" column.

a. Sample calculation of the calculated speed (r = 1.85 × 10^{20} m)

$$v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{(6.67 \times 10^{-11})(1.54 \times 10^{41})}{1.85 \times 10^{20}}} = 2.3563 \times 10^{5} \text{ m/s}$$

b. Sample calculation of the gravitational mass (r = 1.85 × 10^{20} m)

$$M = \frac{v^2 r}{G} = \frac{(2.47 \times 10^{5})^2 (1.85 \times 10^{20})}{6.67 \times 10^{-11}} = 1.6922 \times 10^{41} \text{ kg}$$

c. Sample calculation of the missing mass ($r = 1.85 \times 10^{20}$ m)

$$\text{Missing} = \frac{M - 1.54 \times 10^{41}}{M} \times 100\% = \frac{1.6922 - 1.54}{1.6922} \times 100\% = 9.0\%$$

Completed chart

Orbital radius ($\times 10^{20}$ m)	Calculated speed ($\times 10^{5}$ m/s)	Measured speed ($\times 10^{5}$ m/s)	Gravitational mass ($\times 10^{41}$ kg)	Missing mass (%)
1.85	2.36	2.47	1.69	9.0
2.75	1.93	2.40	2.37	35
3.18	1.80	2.37	2.68	42
4.26	1.55	2.25	3.23	52
6.48	1.26	2.47	5.93	74

FINAL ANSWER

Calculated speeds: 2.36, 1.93, 1.80, 1.55, 1.26 ($\times 10^{5}$ m/s)

Gravitational masses: 1.69, 2.37, 2.68, 3.23, 5.93 ($\times 10^{41}$ kg)

Missing mass: 9.0, 35, 42, 52, 74 %

QUESTIONS 4–6

Plots and what they show

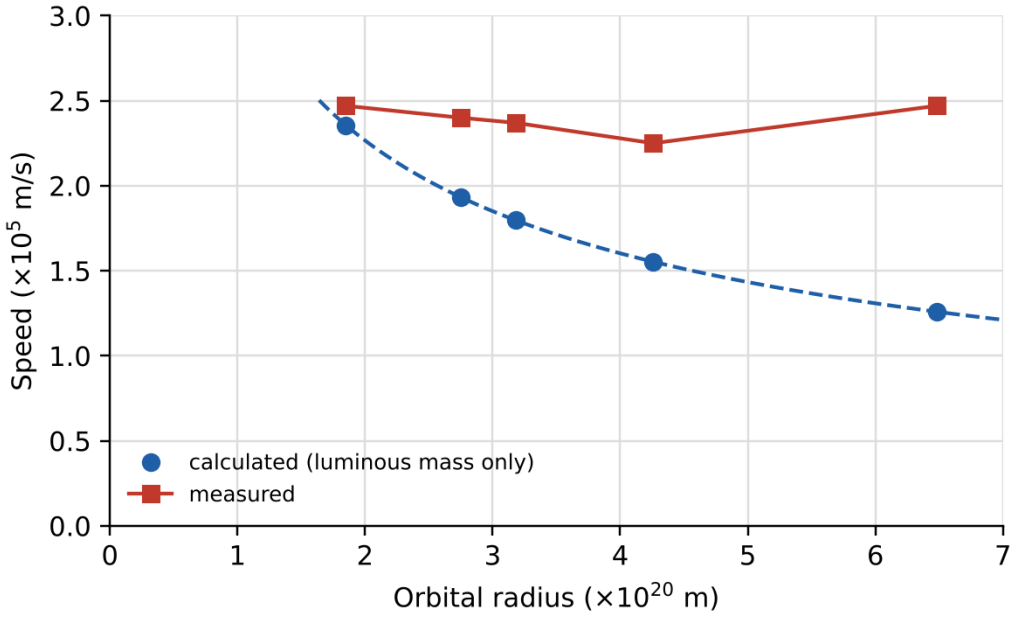
4. a. Plot **calculated** speed versus orbital radius on Logger Pro. Make a sketch below.

   b. Plot **measured** speed against orbital radius on the same axes.

5. Compare the “measured” and “calculated” plots. Explain the difference between the two plots.

6. Explain the shape of your plot for measured speed against orbital radius.

Speed against orbital radius (the dashed line is v = √(GM/r) with the luminous mass)



5. Compare the “measured” and “calculated” plots. Explain the difference.

The calculated speeds fall off with distance ($v \propto 1/\sqrt{r}$), like the planets around the Sun. The measured speeds are higher at every radius and the gap grows farther out. Faster orbits need more centripetal force, so there must be more mass inside each orbit than the stars we can see.

6. Explain the shape of your plot for measured speed against orbital radius.

It is nearly flat (about 2.2–2.5 × 10^5 m/s at every radius). For v = √(GM/r) to stay the same as r grows, the mass inside the orbit must grow in step with r. The gravitational masses in the chart do keep growing even beyond the visible stars, so unseen (dark) matter extends far out into the galaxy.

FINAL ANSWER

5. Measured speeds stay high while calculated speeds drop, so there is extra unseen mass

6. Flat: the mass inside each orbit keeps increasing with r, because dark matter extends far beyond the visible stars